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(1) Research Program/Project/Study Title: Effects of Corn and Mungbean Sprouts as Alternative Nutrient Sources on the Growth Performance of Darag Native Chicken (<i>Gallus gallus Domesticus</i>)	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

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Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☒ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation							
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(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus)
Research:	Development:						
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		☐ Collaborative Research (external agency)
(7) Sector/Commodity Animal		(8) Discipline/Field of Specialization Animal Nutrition
(9A) Abstract		
(9B) Keywords		
(10) Introduction The consumption of chicken meat has experienced exponential growth over the past several decades, making it one of the most widely consumed meats globally. Its popularity is not solely due to its affordability but also to its ability to adapt to diverse culinary traditions and dietary preferences. Its importance as accessible sources of protein, essential vitamins, and minerals for people living in poverty cannot be overstated (FAO, 2013). As global per capita consumption of eggs and white meat continues to rise, the demand for poultry production has surged (Alinaitwe et al., 2018). Chicken farming has emerged as the fastest-growing sector in the livestock industry, and this trend is particularly pronounced in developing countries like the Philippines. However, the soaring demand for poultry production comes with its own set of challenges, most notably the cost of poultry feed. Under intensive production systems, feed expenses can account for a substantial portion of the total production cost, often ranging from 60% to 70%. This financial burden has prompted innovative approaches, such as sprouted fodder production, which is increasingly being adopted in more developed countries (Alinaitwe et al., 2018). Sprouted fodder, in addition to serving as a feed source for livestock during times of fodder scarcity, is also gaining recognition as a suitable poultry feed. Rama et al. (2018) and Chavan and Kadam (1989), noted that seed sprouting undergoes a series of biochemical transformations. These changes encompass the conversion of seed proteins into amino acids, the conversion of carbohydrates into soluble sugars, and the transformation of lipids into essential fatty acids. Furthermore, sprouting triggers an increase in enzyme levels within the seeds, leading to elevated levels of crude protein (CP), ether extract (EE), vitamins, minerals, and improved digestibility. These modifications in nutritional content can have a profound impact on the suitability of sprouted seeds as a nutrient source especially in native chickens that are naturally inclined to thrive on fresh vegetation as their source of nutrients. Energy and protein sources are two fundamental components of animal nutrition, each playing a critical role in the overall health, growth, and productivity of animals. One of the most abundant energy source feedstuff in the Philippines is corn. It is commonly use as primary energy source in animal feed formulations. In a dry matter basis, corn has 3350 kcal/kg metabolizable energy and about 7.5-8% CP. While, the commonly used protein feedstuff is soybean. It is preferred in feed formulations due to its excellent protein content and amino acid profile. However, the production and processing of soybeans can be expensive, contributing to the high cost of soybean-based feeds. This cost can be a significant financial burden for livestock producers, particularly in regions where resources are limited. Recognizing the cost and availability challenges associated with soybean, many feed producers and researchers are exploring alternative protein sources that are more cost-effective and readily available. These alternatives include other commonly available plant protein sources like legumes. The most common legume in the country that can be sprouted is mungbean. The amino acid profile of mungbean sprouts is noteworthy. They contain a range of essential amino acids, which are crucial for animal health and growth. The amino acids in mungbean sprouts can complement those from other feed sources. Despite these claims, there remains a noticeable gap in the available findings, particularly within specific local contexts. It is essential to investigate the ideal germination days of corn that would result in improved		

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growth performance of native chickens specifically, to determine the variation in the nutrient content and percent yield of corn and mungbean sprouts harvested and feed in 3, 6, and 9 days, and its effect in the total feed intake, weight gain, and feed conversion ratio of Darag native chicken; hence, this study is conducted.

(11) Materials and Methods or Methodology

One hundred sixty heads of two-week old Darag native chicken were outsourced locally. They were brood artificially and supplemented with vitamins and mineral in experimental cages until one month of age.

The preparation and germination process for the yellow corn seeds are adopted from Morgan et al. (1992) wherein, local yellow corn seeds, were subjected to an initial cleaning process, by washing it with water to remove impurities and any surface contaminants. Then, the seeds were soaked in a 0.1% hypochlorite solution for 24 hours to disinfects the seeds, reducing the risk of fungal contamination, and promotes the metabolic processes necessary for germination. After the hypochlorite treatment, the seeds were repeatedly washed thoroughly until all impurities and any remaining traces of the hypochlorite solution were removed. Then, the treated seeds were evenly spread out in a 3-layer sprouting deck with a height of 12 inches per deck measuring 118 inches in length by 18 inches width by 3 inches in depth covered with sacks at a seeding rate of seeding rate of 10 kilograms. The sprouts were watered three times daily at 7 am, 12nn, and 4 pm to ensure that the seeds have enough moisture for germination. The sprouting deck was placed under the vacant animal shed covered with mosquito net to prevent the seeds from insect and pests attack.

After 3, 6, and 9 days germination, the sprouts were placed in a net bag and hang to remove extra moisture. It was chopped into approximately 3-5 mm and fed firstly to the birds followed by the commercial feed at a rate of 50% corn sprout and 50% commercial feed.

The acclimatized chicks were randomly assigned to each of the treatments following a Completely Randomized Design (CRD). Germination days (3d, 6d, and 9d) of corn and mungbean seeds as treatments, replicated 4 times with 10 birds per replicate.

The nutritive content of the sprouts was done through proximate analysis at Lipa Quality Control Laboratory. About 100g of oven-dried samples per treatment were grounded and subjected to the analysis of percent DM, Ash, CP, EE and Ca, P, Mg, and K were analyzed using Association of Official Analytical Chemists (AOAC) Official Method of Anaysis, 20th Edition, 2016. While, percent yield was determined by dividing the final weight of the corn sprout biomass over the initial weight of seeds multiplied by 100.

Data were analyzed using one-way analysis of variance (ANOVA) in CRD tested for homogeneity using Bartlett's test and normality using Shapiro's test. Trend comparison was done using Tukeys's Honest Significant Difference Test. Results are statistically significant at the 0.05 alpha level.

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(12) Results and Discussions

Nutrient Composition of Corn and Mungbean Sprouts

Table 1. Chemical composition of corn

Item	Unsprouted seeds	Sprouting day			SEM	P-VALUE
		3d	6d	9d		
DM, %	94.91 ^B	97.76 ^A	97.90 ^A	98.78 ^A	0.39	0.00
CP, %	8.29 ^B	9.08 ^{AB}	8.93 ^{AB}	9.46 ^A	0.15	0.01
EE, %	3.62 ^B	4.27 ^A	3.06 ^C	3.60 ^B	0.11	0.00
Ash, %	1.36 ^B	1.59 ^A	1.28 ^C	1.42 ^B	0.03	0.00
CF, %	2.53 ^B	2.23 ^C	2.77 ^B	3.28 ^A	0.10	0.00

Table 2. Chemical composition of mungbean

Item	Unsprouted seeds	Sprouting day			SEM	P-VALUE
		3d	6d	9d		
DM, %	91.72 ^D	92.69 ^C	94.37 ^B	95.58 ^A	0.39	0.00
CP, %	23.38 ^D	30.39 ^C	33.27 ^B	36.16 ^A	1.23	0.00
EE, %	0.92 ^B	1.32 ^B	1.29 ^B	1.87 ^A	0.89	0.00
Ash, %	3.65 ^D	4.53 ^C	5.40 ^B	6.31 ^A	0.26	0.00
CF, %	4.09 ^D	7.46 ^C	8.46 ^B	13.65 ^A	0.89	0.00

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Table 3. Mineral content of corn and mungbean sprouts

Item	Corn				Mungbean				SEM	P-Value
	Unsprouted	3d	6d	9d	Unsprouted	3d	6d	9d		
Ca, %	0.33 ^A	0.29 ^{AB}	0.28 ^{AB}	0.34 ^A	0.11 ^B	0.29 ^{AB}	0.32 ^A	0.42 ^A	0.02	0.00
P, %	0.24 ^{DE}	0.25 ^{DE}	0.19 ^F	0.22 ^{EF}	0.28 ^D	0.48 ^C	0.54 ^B	0.63 ^A	0.03	0.00
Mg, %	0.56 ^E	0.54 ^{EF}	0.43 ^{FG}	0.38 ^G	1.29 ^D	1.67 ^C	1.95 ^B	2.36 ^A	0.13	0.00
K, %	4.15 ^E	4.14 ^E	3.20 ^E	3.63 ^E	11.66 ^D	13.67 ^C	16.24 ^B	17.66 ^A	1.04	0.00

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Growth Performance of Darag Native Chicken

Yield and Cost Analysis of Corn and Mungbean Sprouts

Herbage Yield

Growth Performance

